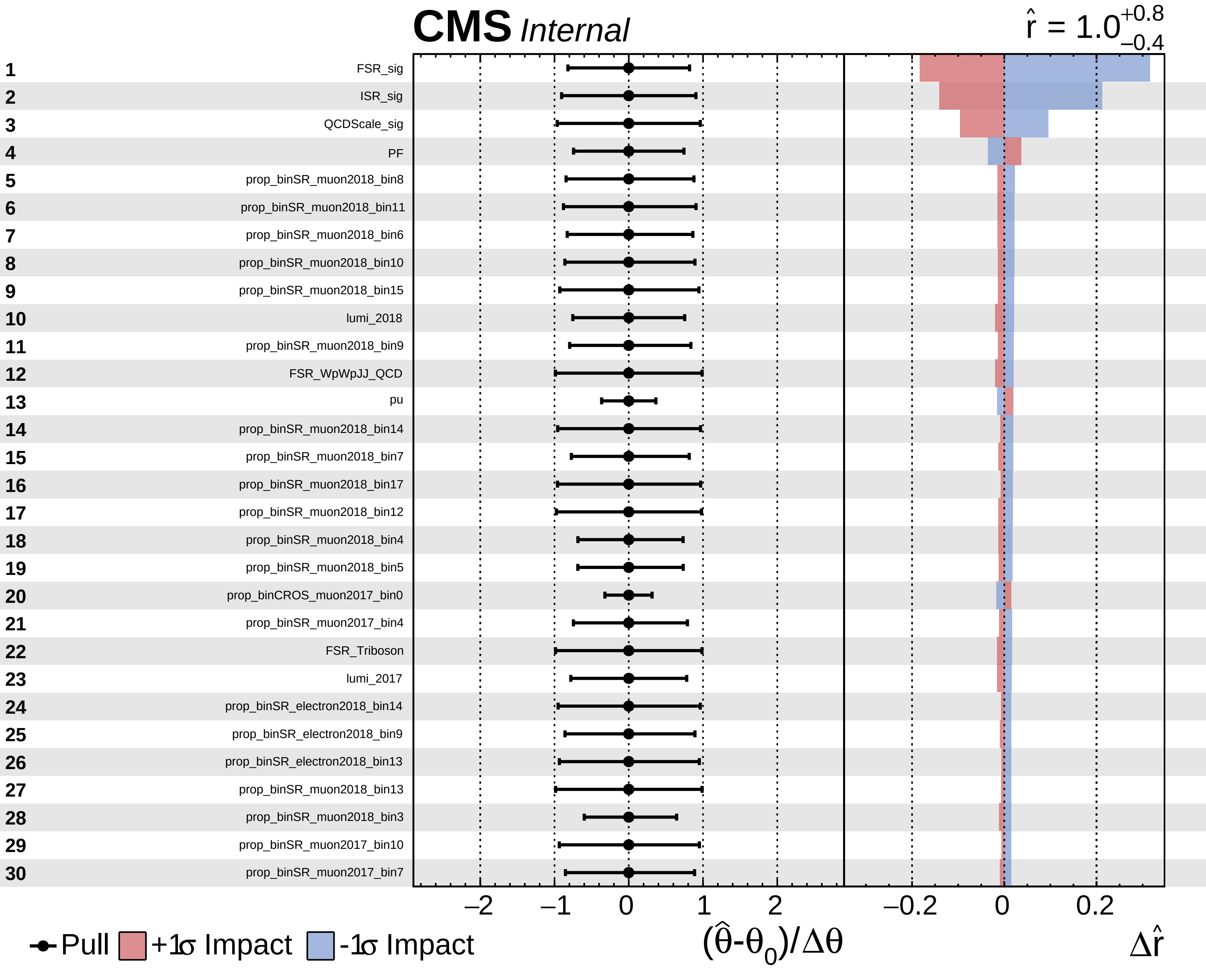
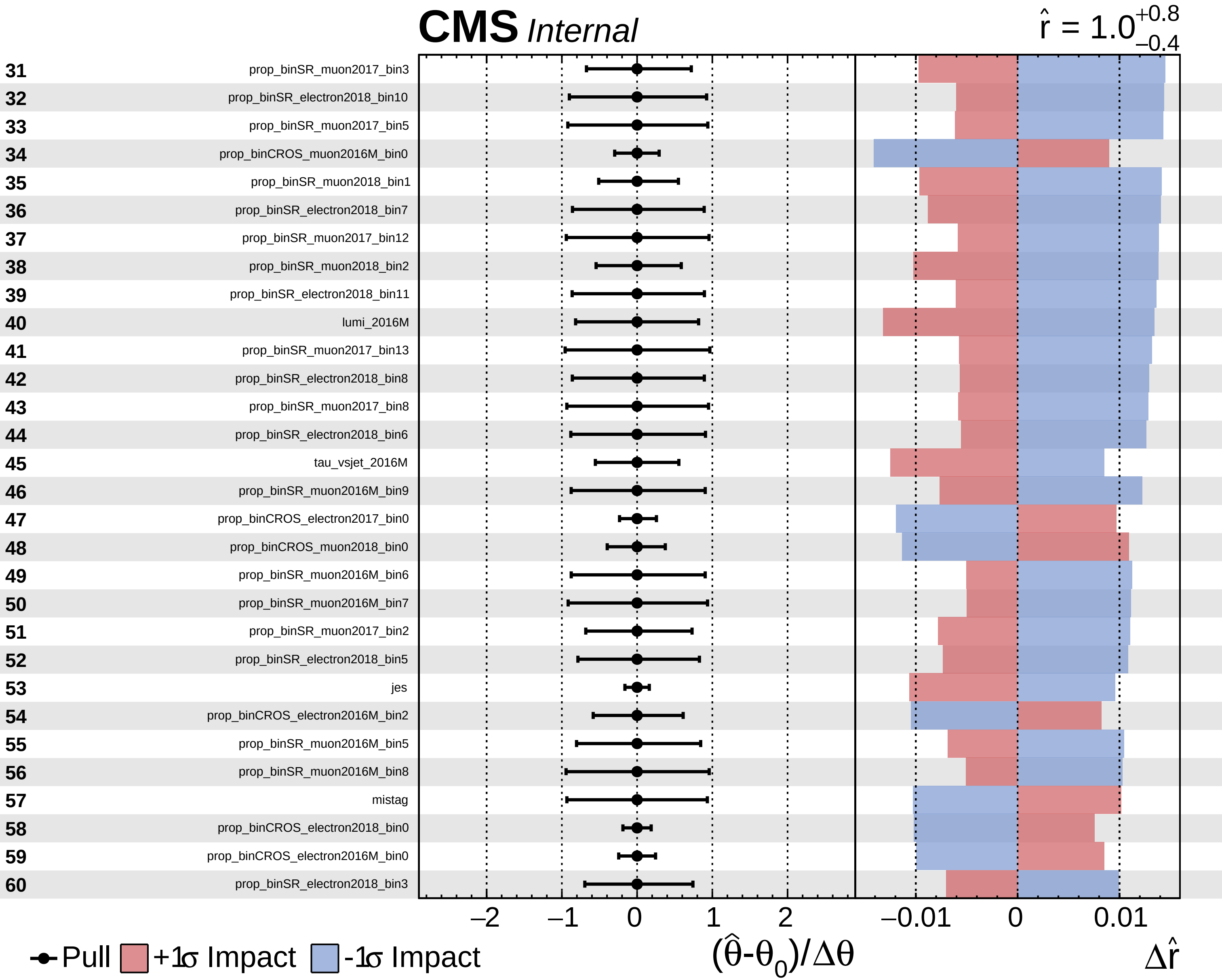
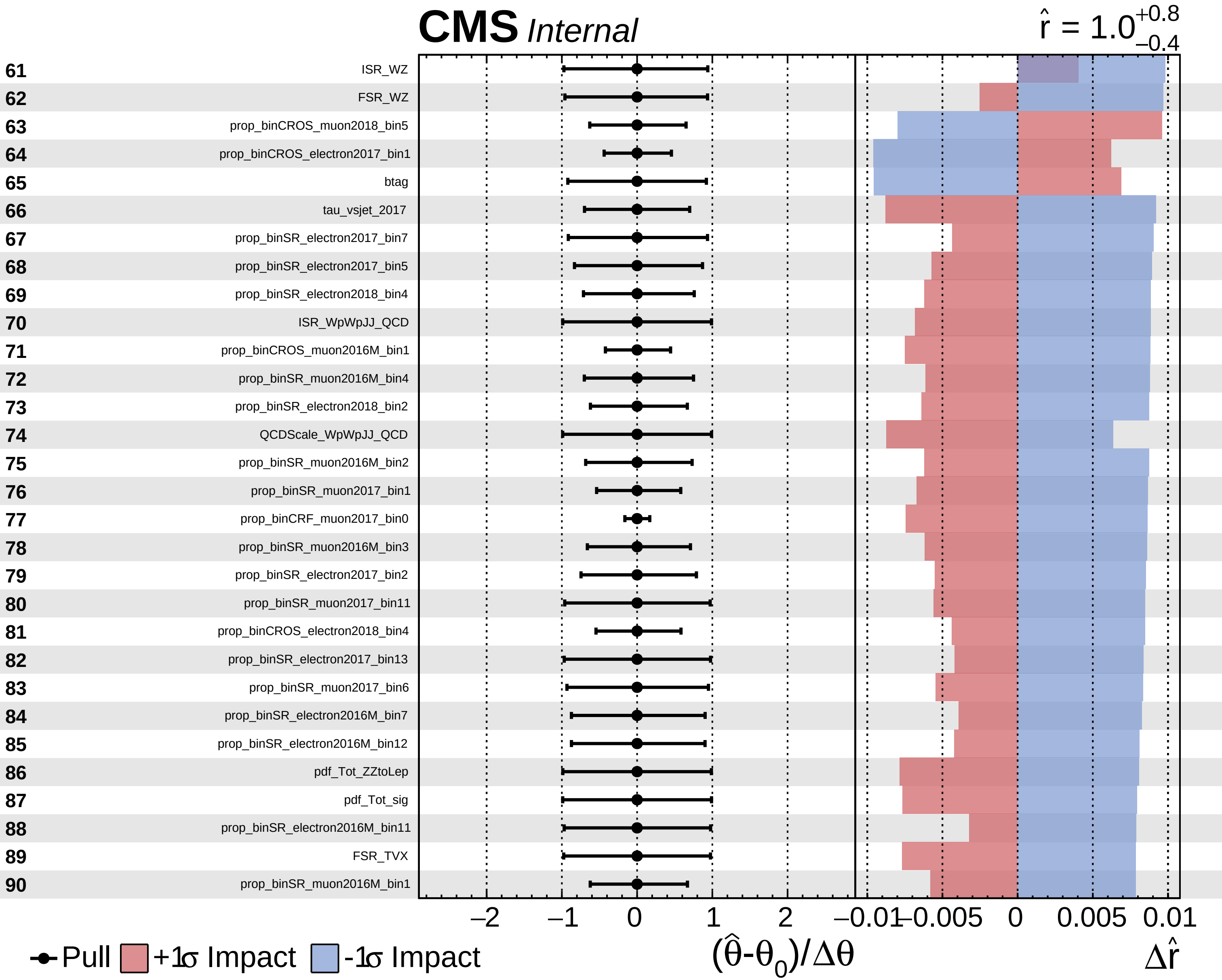
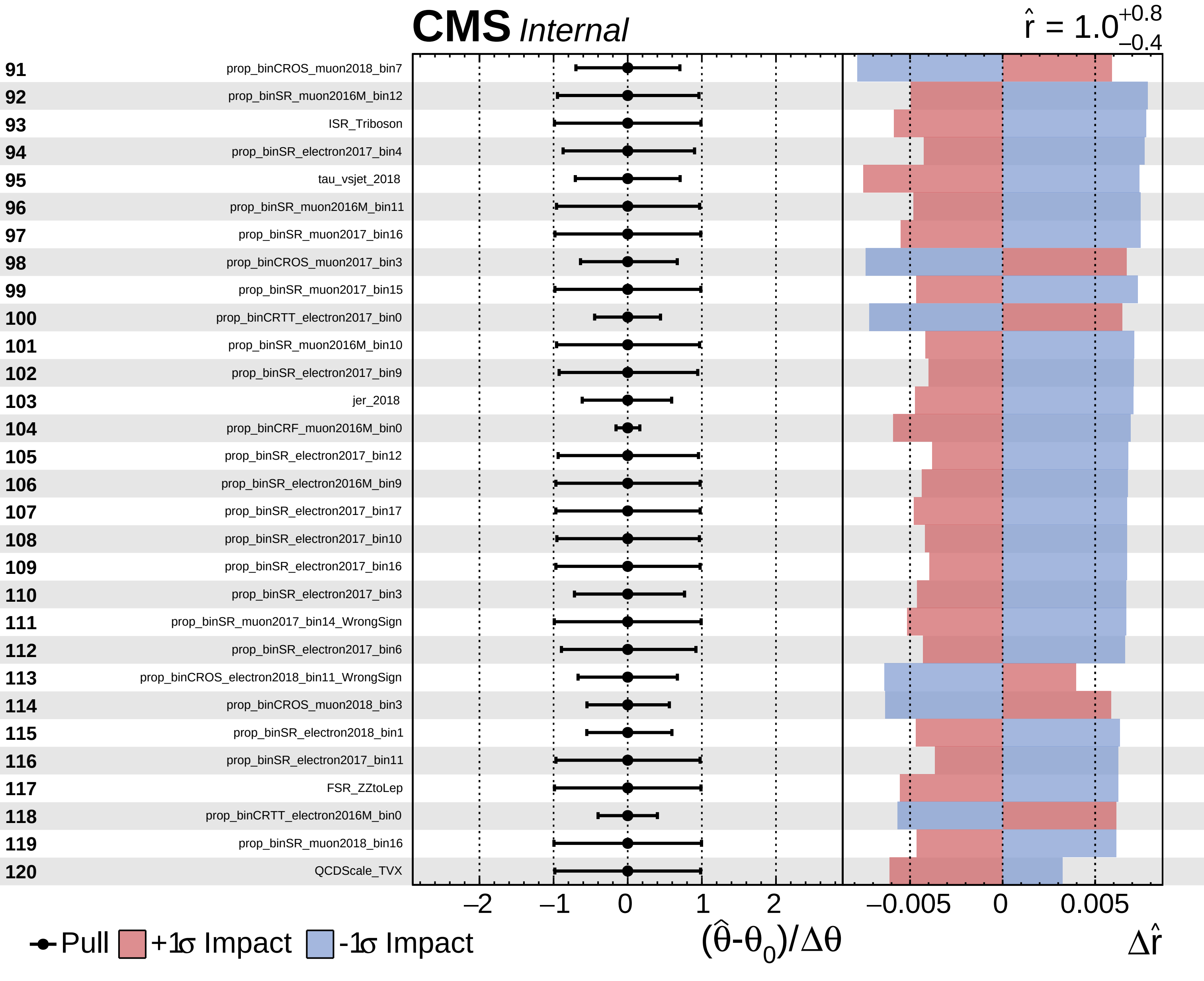


CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
1 FSR_sig
2 ISR_sig
3 QCDScale_sig
4 PF
5 prop_binSR_muon2018_bin8
6 prop_binSR_muon2018_bin11
7 prop_binSR_muon2018_bin6
8 prop_binSR_muon2018_bin10
9 prop_binSR_muon2018_bin15
10 lumi_2018
11 prop_binSR_muon2018_bin9
12 FSR_WpWpJJ_QCD
13 pu
14 prop_binSR_muon2018_bin14
15 prop_binSR_muon2018_bin7
16 prop_binSR_muon2018_bin17
17 prop_binSR_muon2018_bin12
18 prop_binSR_muon2018_bin4
19 prop_binSR_muon2018_bin5
20 prop_binCROS_muon2017_bin0
21 prop_binSR_muon2017_bin4
22 FSR_Triboson
23 lumi_2017
24 prop_binSR_electron2018_bin14
25 prop_binSR_electron2018_bin9
26 prop_binSR_electron2018_bin13
27 prop_binSR_muon2018_bin13
28 prop_binSR_muon2018_bin3
29 prop_binSR_muon2017_bin10
30 prop_binSR_muon2017_bin7
−2 −1 0 1 2
−0.2 0 0.2
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

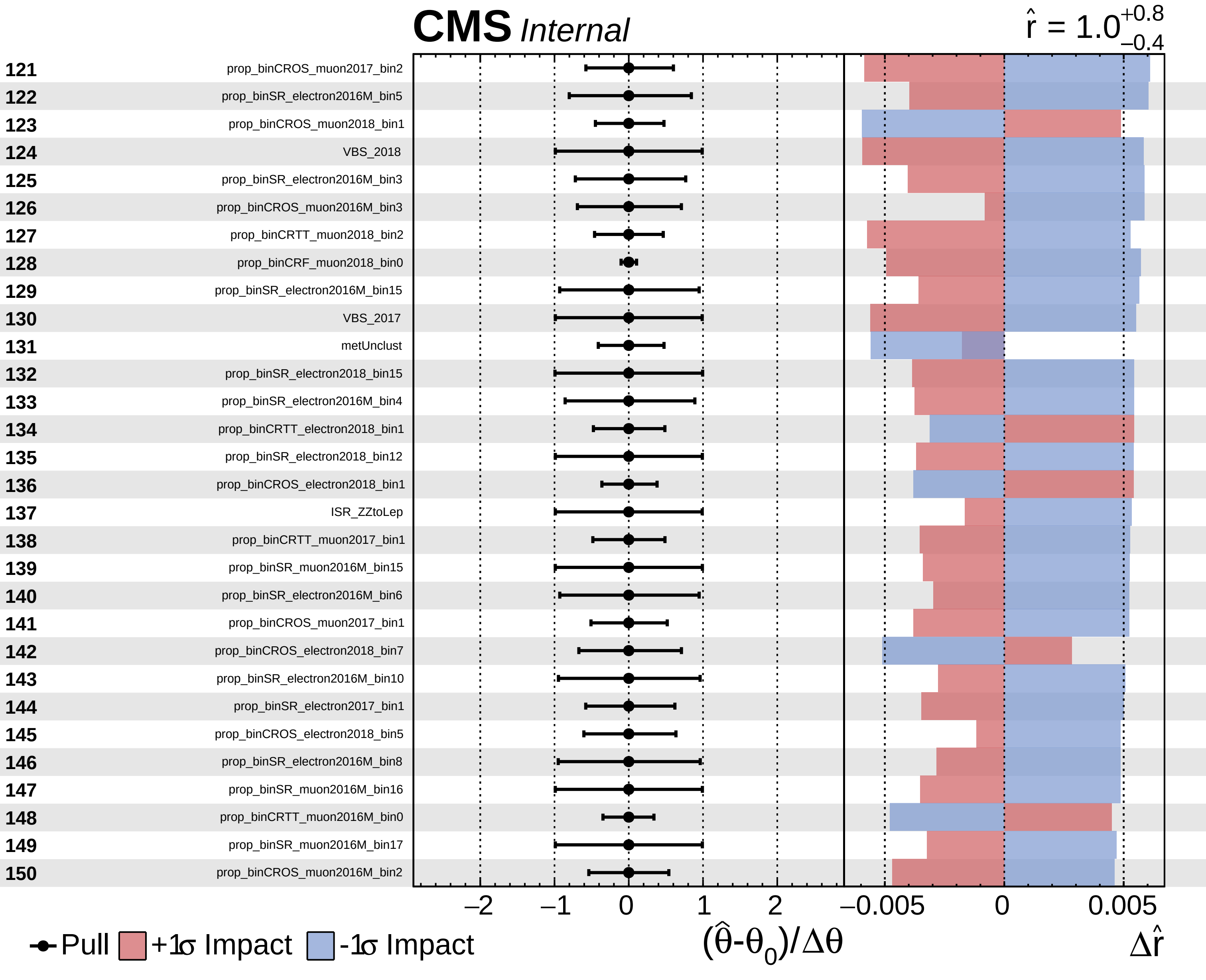

CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
31 prop_binSR_muon2017_bin3
32 prop_binSR_electron2018_bin10
33 prop_binSR_muon2017_bin5
34 prop_binCROS_muon2016M_bin0
35 prop_binSR_muon2018_bin1
36 prop_binSR_electron2018_bin7
37 prop_binSR_muon2017_bin12
38 prop_binSR_muon2018_bin2
39 prop_binSR_electron2018_bin11
40 lumi_2016M
41 prop_binSR_muon2017_bin13
42 prop_binSR_electron2018_bin8
43 prop_binSR_muon2017_bin8
44 prop_binSR_electron2018_bin6
45 tau_vsjet_2016M
46 prop_binSR_muon2016M_bin9
47 prop_binCROS_electron2017_bin0
48 prop_binCROS_muon2018_bin0
49 prop_binSR_muon2016M_bin6
50 prop_binSR_muon2016M_bin7
51 prop_binSR_muon2017_bin2
52 prop_binSR_electron2018_bin5
53 jes
54 prop_binCROS_electron2016M_bin2
55 prop_binSR_muon2016M_bin5
56 prop_binSR_muon2016M_bin8
57 mistag
58 prop_binCROS_electron2018_bin0
59 prop_binCROS_electron2016M_bin0
60 prop_binSR_electron2018_bin3
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

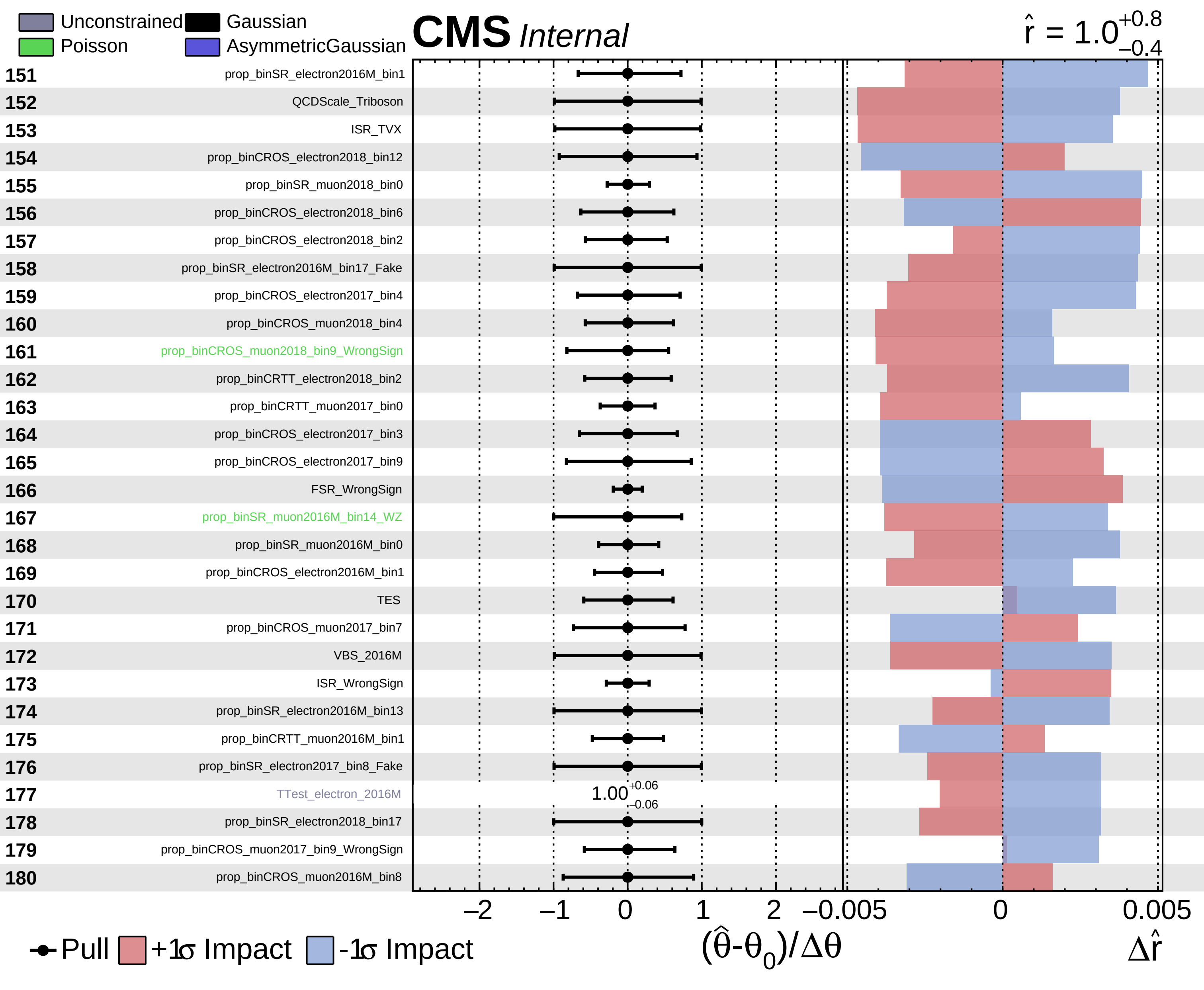


CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
61 ISR_WZ
62 FSR_WZ
63 prop_binCROS_muon2018_bin5
64 prop_binCROS_electron2017_bin1
65 btag
66 tau_vsjet_2017
67 prop_binSR_electron2017_bin7
68 prop_binSR_electron2017_bin5
69 prop_binSR_electron2018_bin4
70 ISR_WpWpJJ_QCD
71 prop_binCROS_muon2016M_bin1
72 prop_binSR_muon2016M_bin4
73 prop_binSR_electron2018_bin2
74 QCDScale_WpWpJJ_QCD
75 prop_binSR_muon2016M_bin2
76 prop_binSR_muon2017_bin1
77 prop_binCRF_muon2017_bin0
78 prop_binSR_muon2016M_bin3
79 prop_binSR_electron2017_bin2
80 prop_binSR_muon2017_bin11
81 prop_binCROS_electron2018_bin4
82 prop_binSR_electron2017_bin13
83 prop_binSR_muon2017_bin6
84 prop_binSR_electron2016M_bin7
85 prop_binSR_electron2016M_bin12
86 pdf_Tot_ZZtoLep
87 pdf_Tot_sig
88 prop_binSR_electron2016M_bin11
89 FSR_TVX
90 prop_binSR_muon2016M_bin1
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

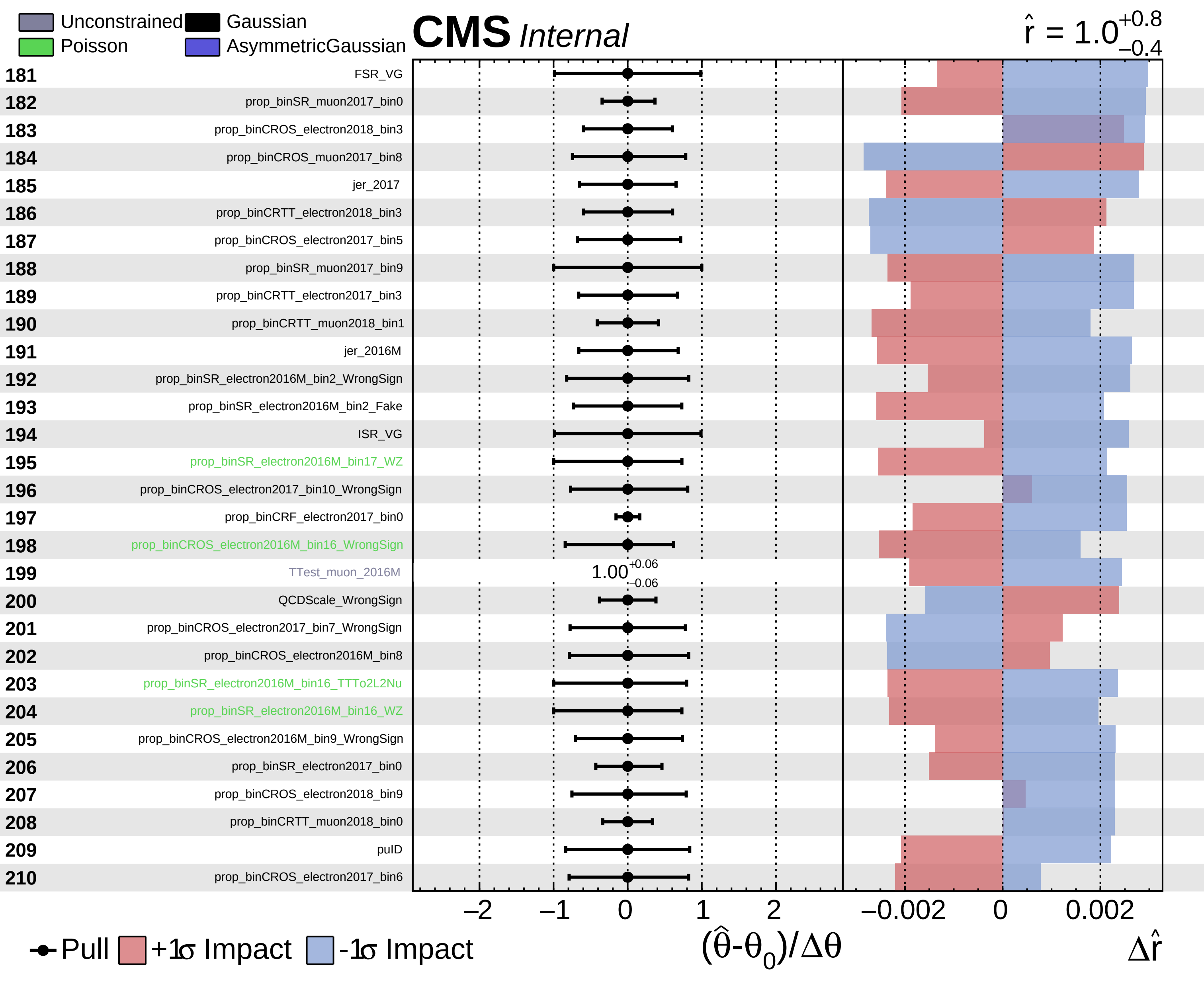


CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
91 prop_binCROS_muon2018_bin7
92 prop_binSR_muon2016M_bin12
93 ISR_Triboson
94 prop_binSR_electron2017_bin4
95 tau_vsjet_2018
96 prop_binSR_muon2016M_bin11
97 prop_binSR_muon2017_bin16
98 prop_binCROS_muon2017_bin3
99 prop_binSR_muon2017_bin15
100 prop_binCRTT_electron2017_bin0
101 prop_binSR_muon2016M_bin10
102 prop_binSR_electron2017_bin9
103 jer_2018
104 prop_binCRF_muon2016M_bin0
105 prop_binSR_electron2017_bin12
106 prop_binSR_electron2016M_bin9
107 prop_binSR_electron2017_bin17
108 prop_binSR_electron2017_bin10
109 prop_binSR_electron2017_bin16
110 prop_binSR_electron2017_bin3
111 prop_binSR_muon2017_bin14_WrongSign
112 prop_binSR_electron2017_bin6
113 prop_binCROS_electron2018_bin11_WrongSign
114 prop_binCROS_muon2018_bin3
115 prop_binSR_electron2018_bin1
116 prop_binSR_electron2017_bin11
117 FSR_ZZtoLep
118 prop_binCRTT_electron2016M_bin0
119 prop_binSR_muon2018_bin16
120 QCDScale_TVX
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

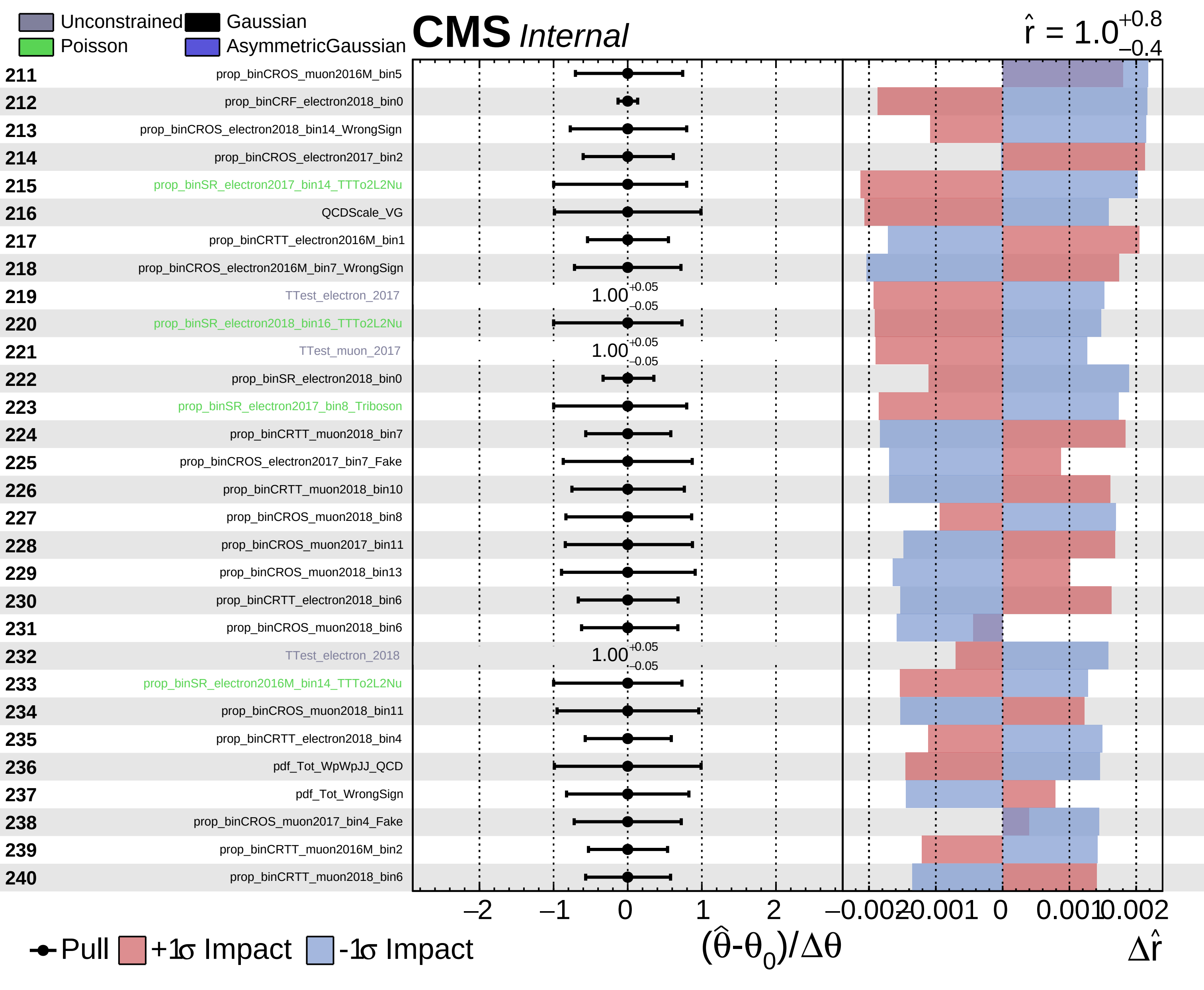

CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
121 prop_binCROS_muon2017_bin2
122 prop_binSR_electron2016M_bin5
123 prop_binCROS_muon2018_bin1
124 VBS_2018
125 prop_binSR_electron2016M_bin3
126 prop_binCROS_muon2016M_bin3
127 prop_binCRTT_muon2018_bin2
128 prop_binCRF_muon2018_bin0
129 prop_binSR_electron2016M_bin15
130 VBS_2017
131 metUnclust
132 prop_binSR_electron2018_bin15
133 prop_binSR_electron2016M_bin4
134 prop_binCRTT_electron2018_bin1
135 prop_binSR_electron2018_bin12
136 prop_binCROS_electron2018_bin1
137 ISR_ZZtoLep
138 prop_binCRTT_muon2017_bin1
139 prop_binSR_muon2016M_bin15
140 prop_binSR_electron2016M_bin6
141 prop_binCROS_muon2017_bin1
142 prop_binCROS_electron2018_bin7
143 prop_binSR_electron2016M_bin10
144 prop_binSR_electron2017_bin1
145 prop_binCROS_electron2018_bin5
146 prop_binSR_electron2016M_bin8
147 prop_binSR_muon2016M_bin16
148 prop_binCRTT_muon2016M_bin0
149 prop_binSR_muon2016M_bin17
150 prop_binCROS_muon2016M_bin2
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



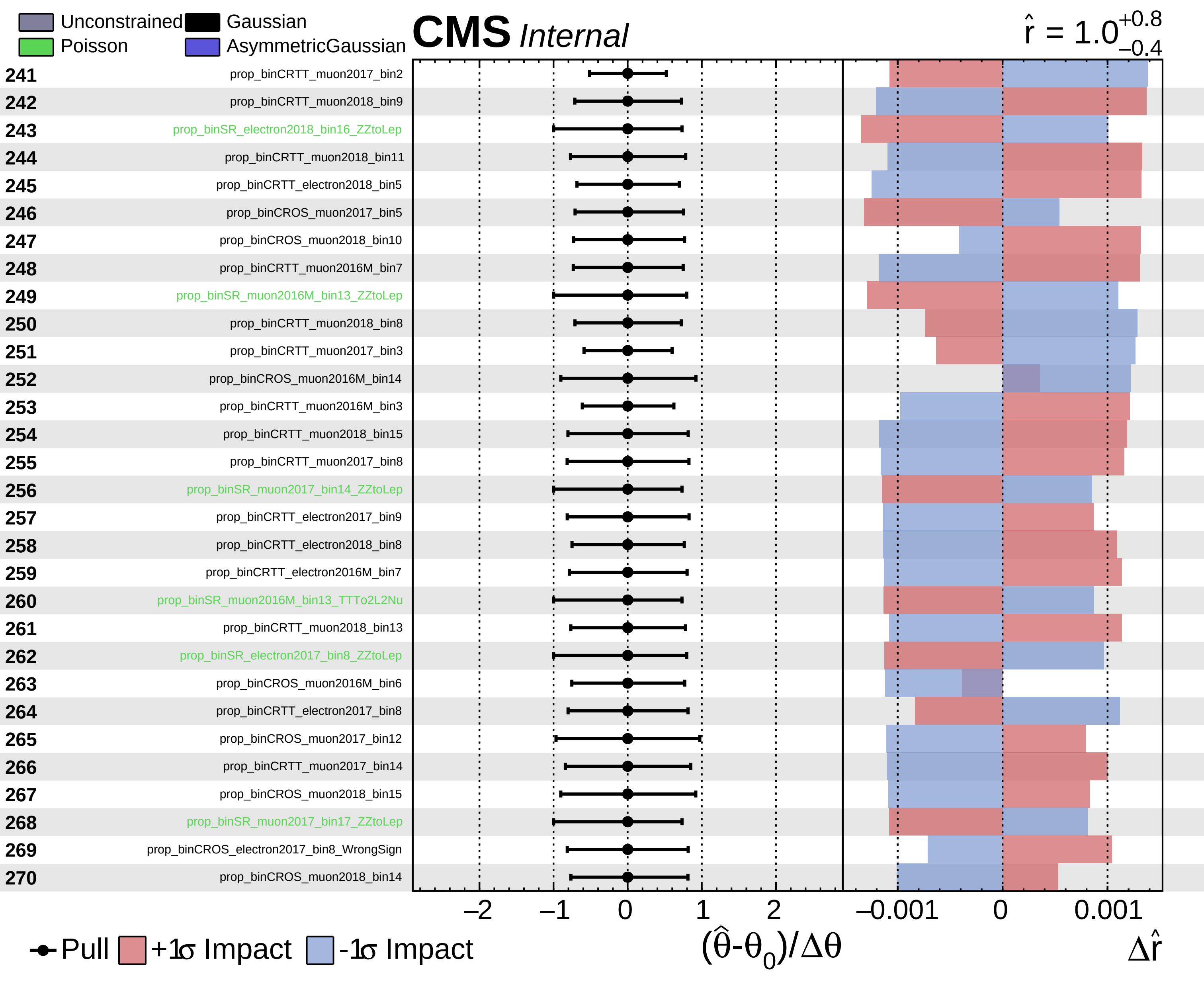
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
151 prop_binSR_electron2016M_bin1
152 QCDScale_Triboson
153 ISR_TVX
154 prop_binCROS_electron2018_bin12
155 prop_binSR_muon2018_bin0
156 prop_binCROS_electron2018_bin6
157 prop_binCROS_electron2018_bin2
158 prop_binSR_electron2016M_bin17_Fake
159 prop_binCROS_electron2017_bin4
160 prop_binCROS_muon2018_bin4
161 prop_binCROS_muon2018_bin9_WrongSign
162 prop_binCRTT_electron2018_bin2
163 prop_binCRTT_muon2017_bin0
164 prop_binCROS_electron2017_bin3
165 prop_binCROS_electron2017_bin9
166 FSR_WrongSign
167 prop_binSR_muon2016M_bin14_WZ
168 prop_binSR_muon2016M_bin0
169 prop_binCROS_electron2016M_bin1
170 TES
171 prop_binCROS_muon2017_bin7
172 VBS_2016M
173 ISR_WrongSign
174 prop_binSR_electron2016M_bin13
175 prop_binCRTT_muon2016M_bin1
176 prop_binSR_electron2017_bin8_Fake
177 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
178 prop_binSR_electron2018_bin17
179 prop_binCROS_muon2017_bin9_WrongSign
180 prop_binCROS_muon2016M_bin8
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



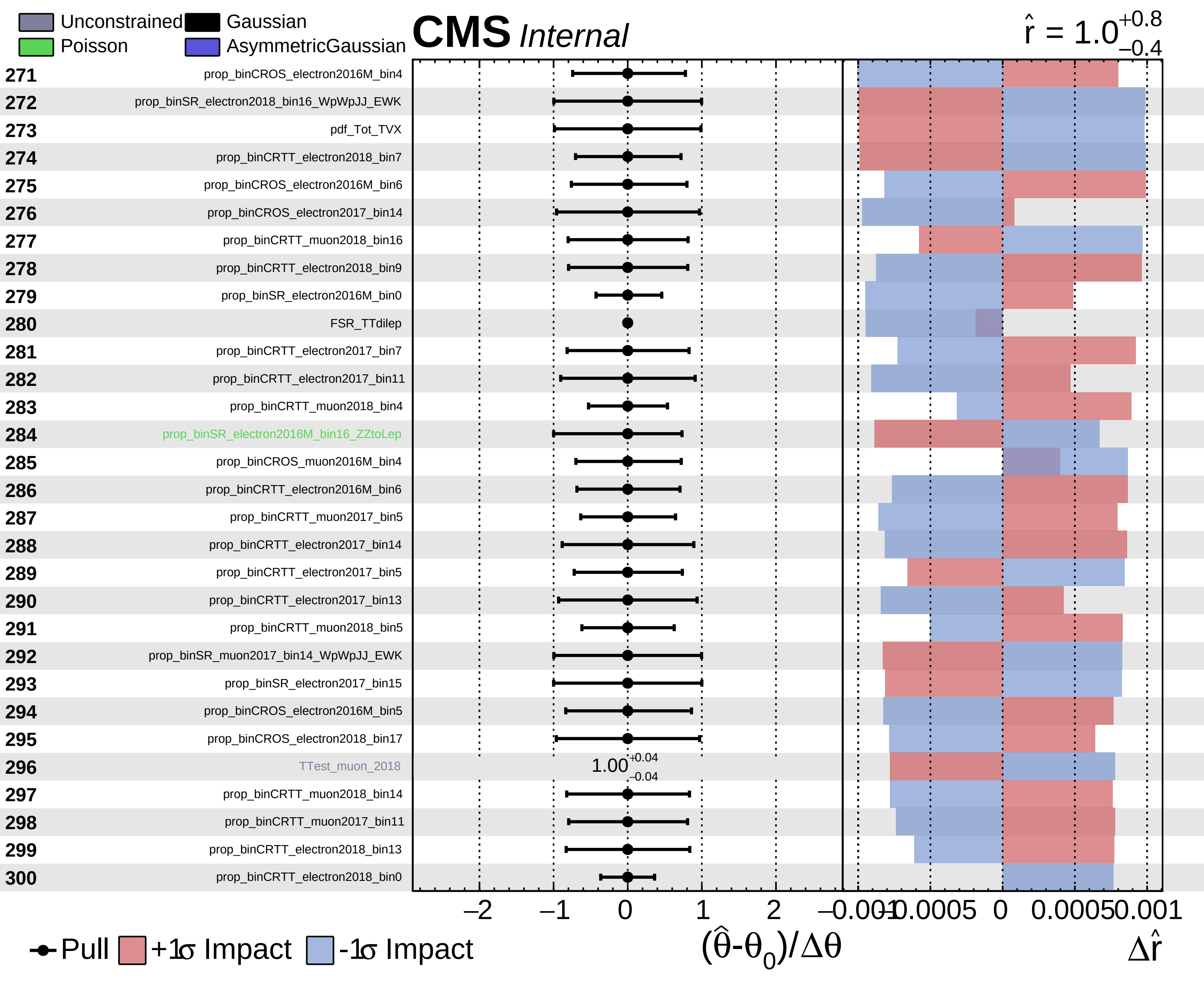
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
181 FSR_VG
182 prop_binSR_muon2017_bin0
183 prop_binCROS_electron2018_bin3
184 prop_binCROS_muon2017_bin8
185 jer_2017
186 prop_binCRTT_electron2018_bin3
187 prop_binCROS_electron2017_bin5
188 prop_binSR_muon2017_bin9
189 prop_binCRTT_electron2017_bin3
190 prop_binCRTT_muon2018_bin1
191 jer_2016M
192 prop_binSR_electron2016M_bin2_WrongSign
193 prop_binSR_electron2016M_bin2_Fake
194 ISR_VG
195 prop_binSR_electron2016M_bin17_WZ
196 prop_binCROS_electron2017_bin10_WrongSign
197 prop_binCRF_electron2017_bin0
198 prop_binCROS_electron2016M_bin16_WrongSign
199 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
200 QCDScale_WrongSign
201 prop_binCROS_electron2017_bin7_WrongSign
202 prop_binCROS_electron2016M_bin8
203 prop_binSR_electron2016M_bin16_TTTo2L2Nu
204 prop_binSR_electron2016M_bin16_WZ
205 prop_binCROS_electron2016M_bin9_WrongSign
206 prop_binSR_electron2017_bin0
207 prop_binCROS_electron2018_bin9
208 prop_binCRTT_muon2018_bin0
209 puID
210 prop_binCROS_electron2017_bin6
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



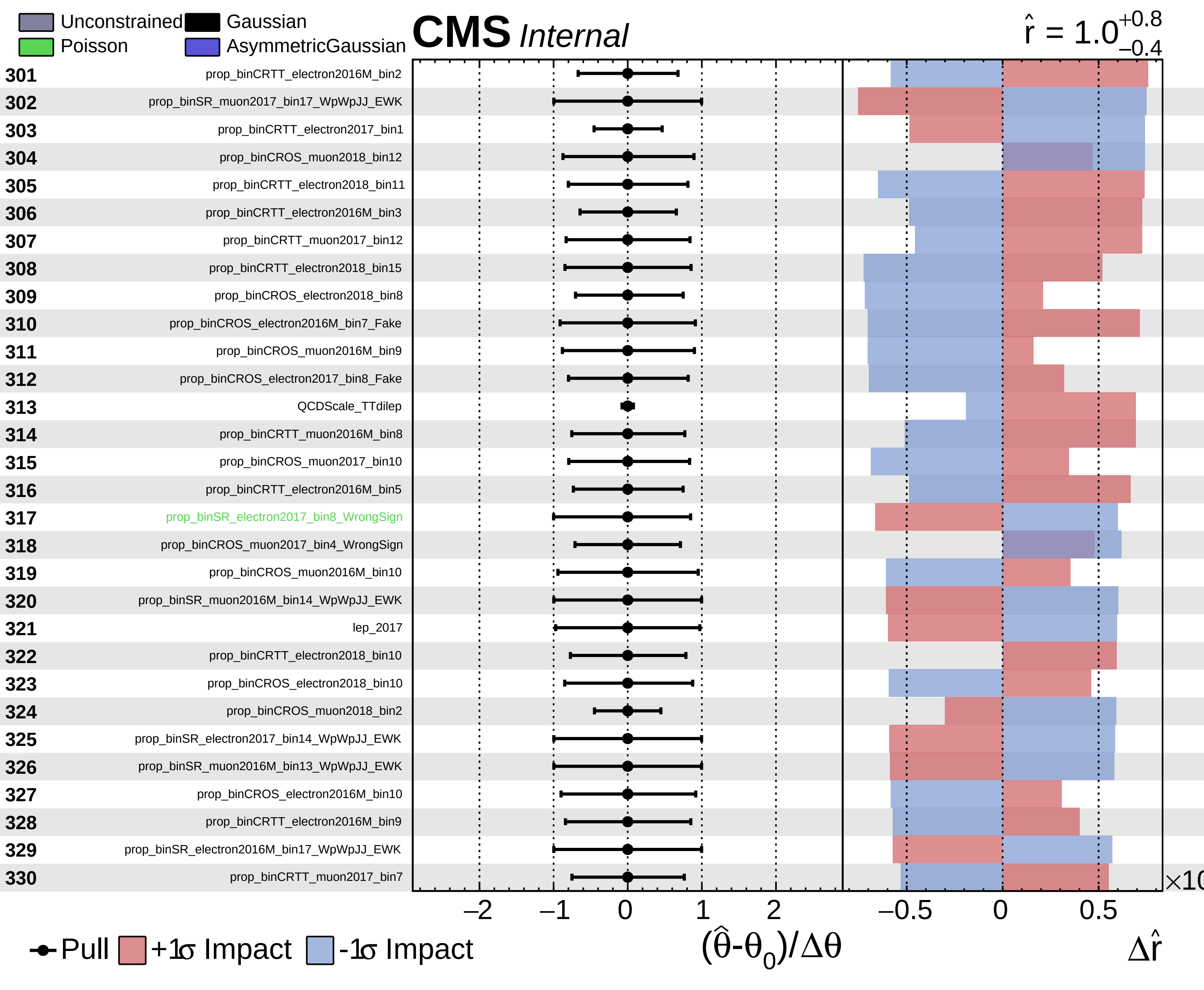
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
211 prop_binCROS_muon2016M_bin5
212 prop_binCRF_electron2018_bin0
213 prop_binCROS_electron2018_bin14_WrongSign
214 prop_binCROS_electron2017_bin2
215 prop_binSR_electron2017_bin14_TTTo2L2Nu
216 QCDScale_VG
217 prop_binCRTT_electron2016M_bin1
218 prop_binCROS_electron2016M_bin7_WrongSign
219 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
220 prop_binSR_electron2018_bin16_TTTo2L2Nu
221 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
222 prop_binSR_electron2018_bin0
223 prop_binSR_electron2017_bin8_Triboson
224 prop_binCRTT_muon2018_bin7
225 prop_binCROS_electron2017_bin7_Fake
226 prop_binCRTT_muon2018_bin10
227 prop_binCROS_muon2018_bin8
228 prop_binCROS_muon2017_bin11
229 prop_binCROS_muon2018_bin13
230 prop_binCRTT_electron2018_bin6
231 prop_binCROS_muon2018_bin6
232 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
233 prop_binSR_electron2016M_bin14_TTTo2L2Nu
234 prop_binCROS_muon2018_bin11
235 prop_binCRTT_electron2018_bin4
236 pdf_Tot_WpWpJJ_QCD
237 pdf_Tot_WrongSign
238 prop_binCROS_muon2017_bin4_Fake
239 prop_binCRTT_muon2016M_bin2
240 prop_binCRTT_muon2018_bin6
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



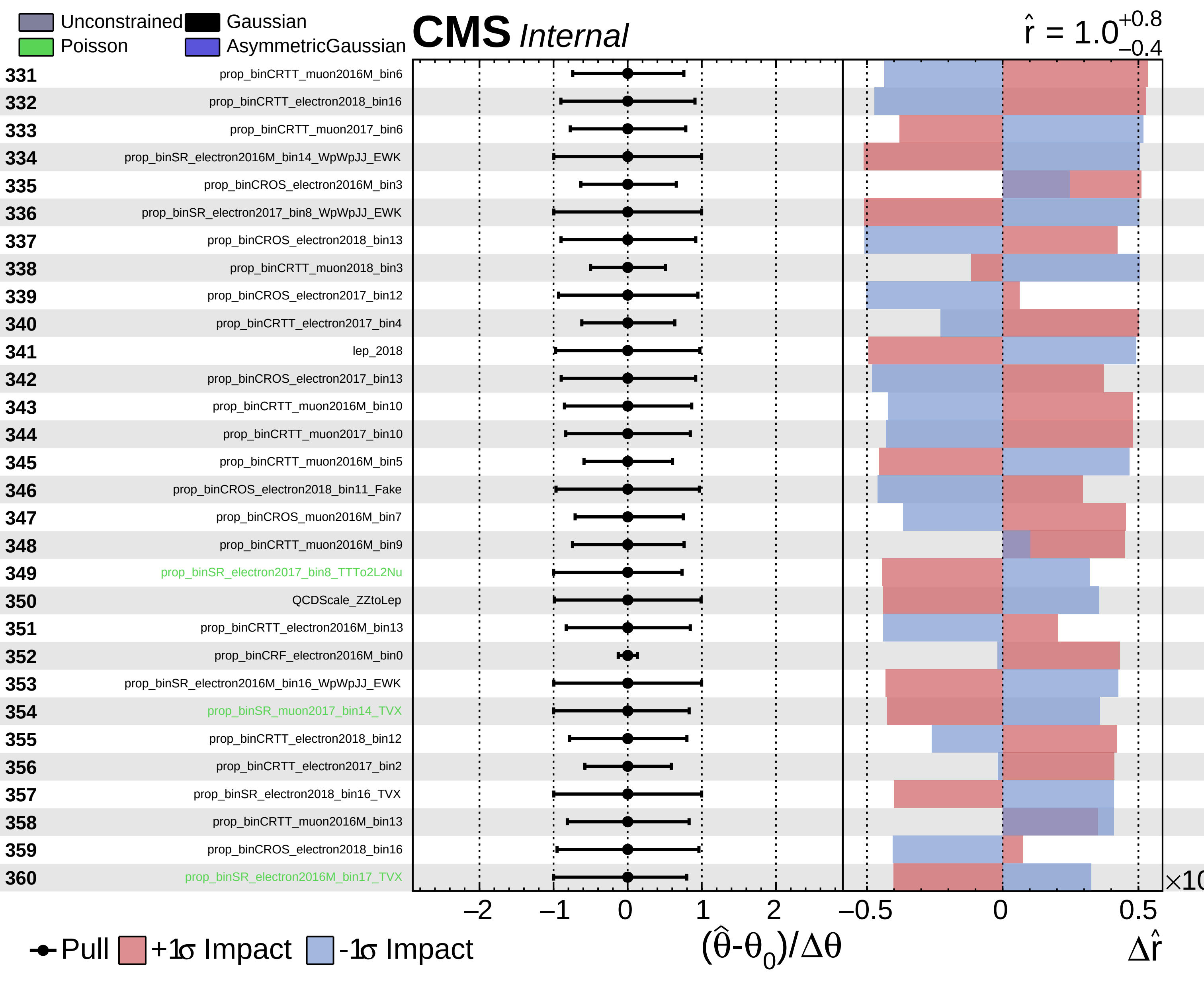
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
241 prop_binCRTT_muon2017_bin2
242 prop_binCRTT_muon2018_bin9
243 prop_binSR_electron2018_bin16_ZZtoLep
244 prop_binCRTT_muon2018_bin11
245 prop_binCRTT_electron2018_bin5
246 prop_binCROS_muon2017_bin5
247 prop_binCROS_muon2018_bin10
248 prop_binCRTT_muon2016M_bin7
249 prop_binSR_muon2016M_bin13_ZZtoLep
250 prop_binCRTT_muon2018_bin8
251 prop_binCRTT_muon2017_bin3
252 prop_binCROS_muon2016M_bin14
253 prop_binCRTT_muon2016M_bin3
254 prop_binCRTT_muon2018_bin15
255 prop_binCRTT_muon2017_bin8
256 prop_binSR_muon2017_bin14_ZZtoLep
257 prop_binCRTT_electron2017_bin9
258 prop_binCRTT_electron2018_bin8
259 prop_binCRTT_electron2016M_bin7
260 prop_binSR_muon2016M_bin13_TTTo2L2Nu
261 prop_binCRTT_muon2018_bin13
262 prop_binSR_electron2017_bin8_ZZtoLep
263 prop_binCROS_muon2016M_bin6
264 prop_binCRTT_electron2017_bin8
265 prop_binCROS_muon2017_bin12
266 prop_binCRTT_muon2017_bin14
267 prop_binCROS_muon2018_bin15
268 prop_binSR_muon2017_bin17_ZZtoLep
269 prop_binCROS_electron2017_bin8_WrongSign
270 prop_binCROS_muon2018_bin14
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



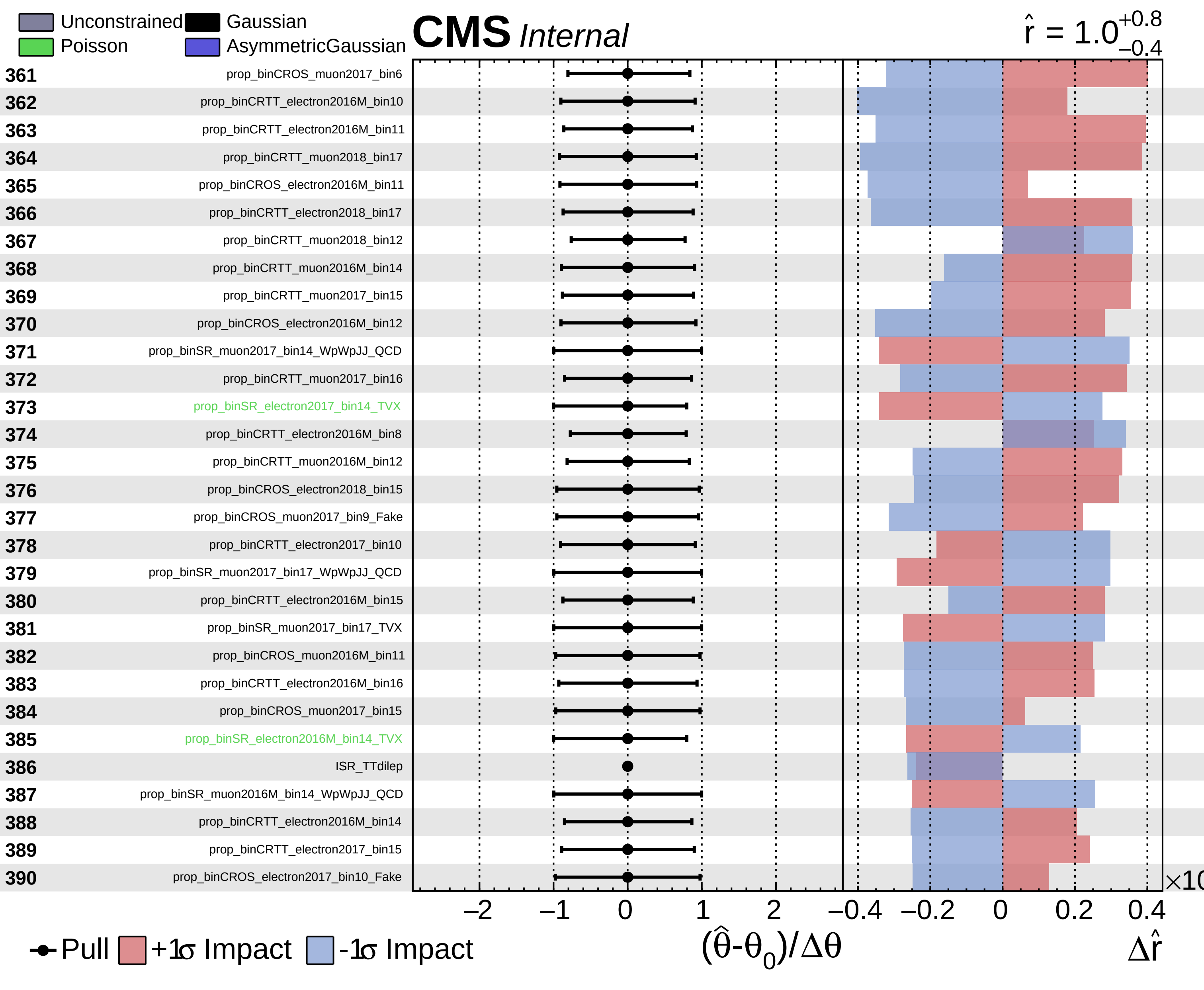
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
271 prop_binCROS_electron2016M_bin4
272 prop_binSR_electron2018_bin16_WpWpJJ_EWK
273 pdf_Tot_TVX
274 prop_binCRTT_electron2018_bin7
275 prop_binCROS_electron2016M_bin6
276 prop_binCROS_electron2017_bin14
277 prop_binCRTT_muon2018_bin16
278 prop_binCRTT_electron2018_bin9
279 prop_binSR_electron2016M_bin0
280 FSR_TTdilep
281 prop_binCRTT_electron2017_bin7
282 prop_binCRTT_electron2017_bin11
283 prop_binCRTT_muon2018_bin4
284 prop_binSR_electron2016M_bin16_ZZtoLep
285 prop_binCROS_muon2016M_bin4
286 prop_binCRTT_electron2016M_bin6
287 prop_binCRTT_muon2017_bin5
288 prop_binCRTT_electron2017_bin14
289 prop_binCRTT_electron2017_bin5
290 prop_binCRTT_electron2017_bin13
291 prop_binCRTT_muon2018_bin5
292 prop_binSR_muon2017_bin14_WpWpJJ_EWK
293 prop_binSR_electron2017_bin15
294 prop_binCROS_electron2016M_bin5
295 prop_binCROS_electron2018_bin17
296 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
297 prop_binCRTT_muon2018_bin14
298 prop_binCRTT_muon2017_bin11
299 prop_binCRTT_electron2018_bin13
300 prop_binCRTT_electron2018_bin0
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



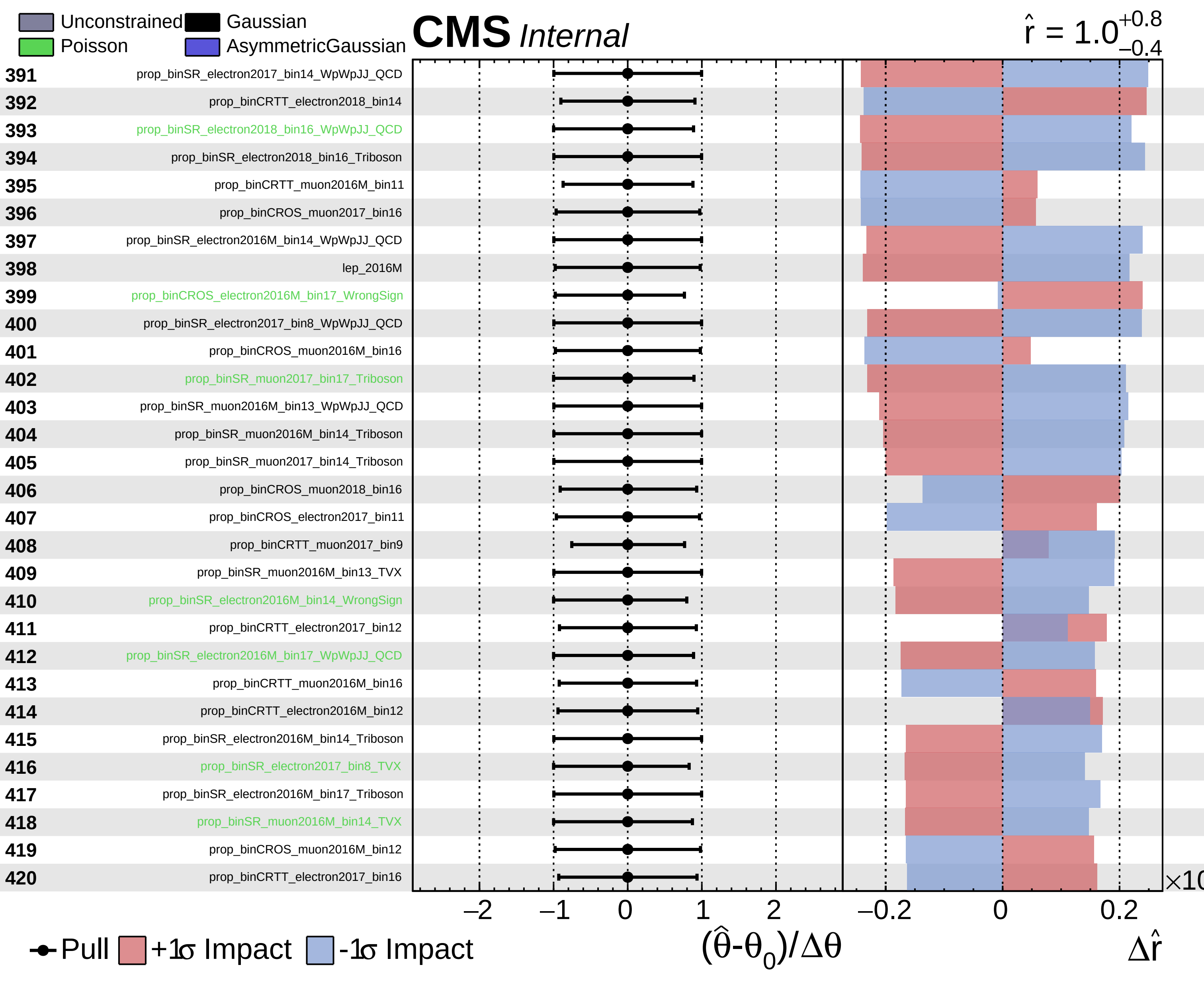
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
301 prop_binCRTT_electron2016M_bin2
302 prop_binSR_muon2017_bin17_WpWpJJ_EWK
303 prop_binCRTT_electron2017_bin1
304 prop_binCROS_muon2018_bin12
305 prop_binCRTT_electron2018_bin11
306 prop_binCRTT_electron2016M_bin3
307 prop_binCRTT_muon2017_bin12
308 prop_binCRTT_electron2018_bin15
309 prop_binCROS_electron2018_bin8
310 prop_binCROS_electron2016M_bin7_Fake
311 prop_binCROS_muon2016M_bin9
312 prop_binCROS_electron2017_bin8_Fake
313 QCDScale_TTdilep
314 prop_binCRTT_muon2016M_bin8
315 prop_binCROS_muon2017_bin10
316 prop_binCRTT_electron2016M_bin5
317 prop_binSR_electron2017_bin8_WrongSign
318 prop_binCROS_muon2017_bin4_WrongSign
319 prop_binCROS_muon2016M_bin10
320 prop_binSR_muon2016M_bin14_WpWpJJ_EWK
321 lep_2017
322 prop_binCRTT_electron2018_bin10
323 prop_binCROS_electron2018_bin10
324 prop_binCROS_muon2018_bin2
325 prop_binSR_electron2017_bin14_WpWpJJ_EWK
326 prop_binSR_muon2016M_bin13_WpWpJJ_EWK
327 prop_binCROS_electron2016M_bin10
328 prop_binCRTT_electron2016M_bin9
329 prop_binSR_electron2016M_bin17_WpWpJJ_EWK
330 prop_binCRTT_muon2017_bin7
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

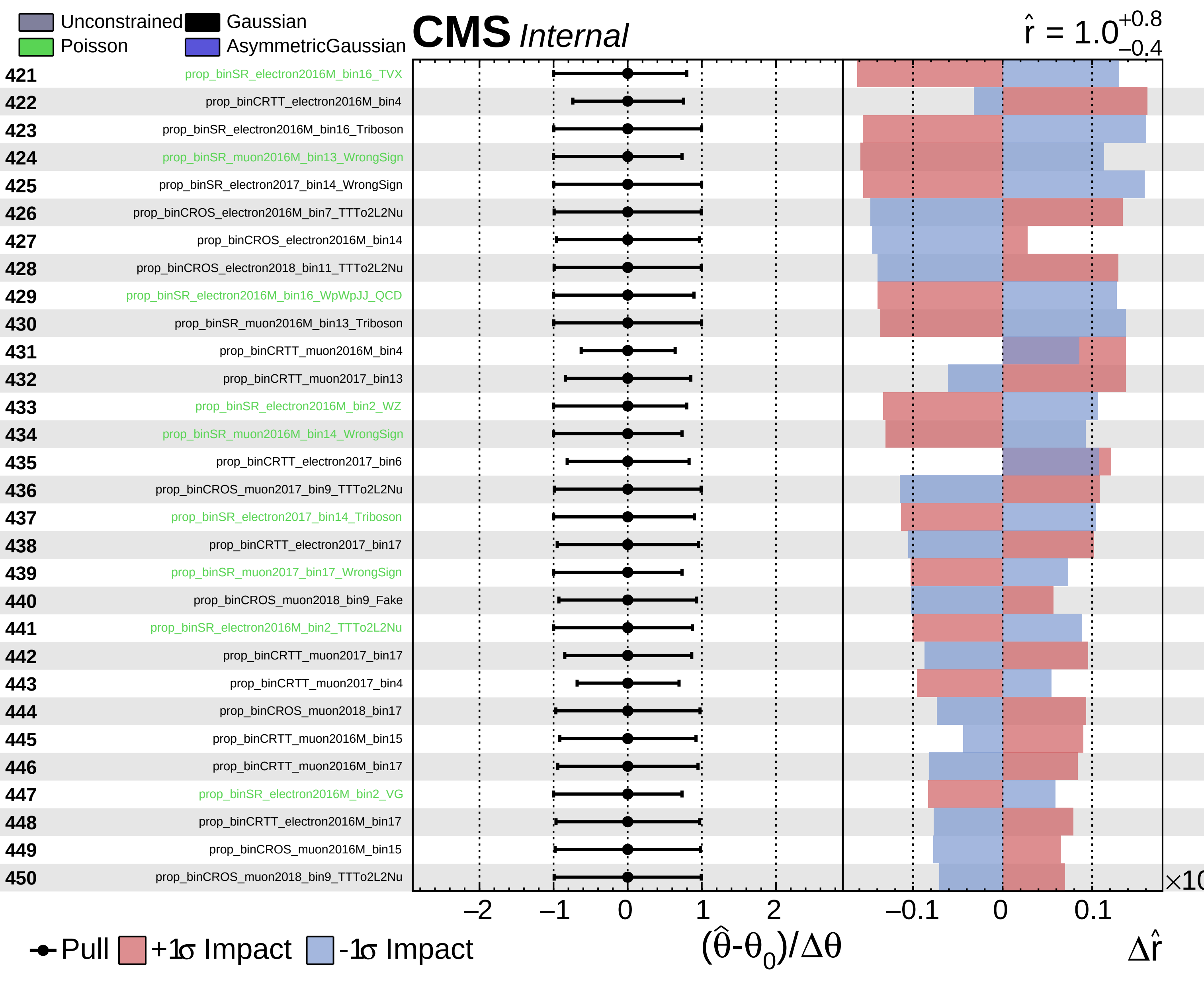


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
331 prop_binCRTT_muon2016M_bin6
332 prop_binCRTT_electron2018_bin16
333 prop_binCRTT_muon2017_bin6
334 prop_binSR_electron2016M_bin14_WpWpJJ_EWK
335 prop_binCROS_electron2016M_bin3
336 prop_binSR_electron2017_bin8_WpWpJJ_EWK
337 prop_binCROS_electron2018_bin13
338 prop_binCRTT_muon2018_bin3
339 prop_binCROS_electron2017_bin12
340 prop_binCRTT_electron2017_bin4
341 lep_2018
342 prop_binCROS_electron2017_bin13
343 prop_binCRTT_muon2016M_bin10
344 prop_binCRTT_muon2017_bin10
345 prop_binCRTT_muon2016M_bin5
346 prop_binCROS_electron2018_bin11_Fake
347 prop_binCROS_muon2016M_bin7
348 prop_binCRTT_muon2016M_bin9
349 prop_binSR_electron2017_bin8_TTTo2L2Nu
350 QCDScale_ZZtoLep
351 prop_binCRTT_electron2016M_bin13
352 prop_binCRF_electron2016M_bin0
353 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
354 prop_binSR_muon2017_bin14_TVX
355 prop_binCRTT_electron2018_bin12
356 prop_binCRTT_electron2017_bin2
357 prop_binSR_electron2018_bin16_TVX
358 prop_binCRTT_muon2016M_bin13
359 prop_binCROS_electron2018_bin16
360 prop_binSR_electron2016M_bin17_TVX
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

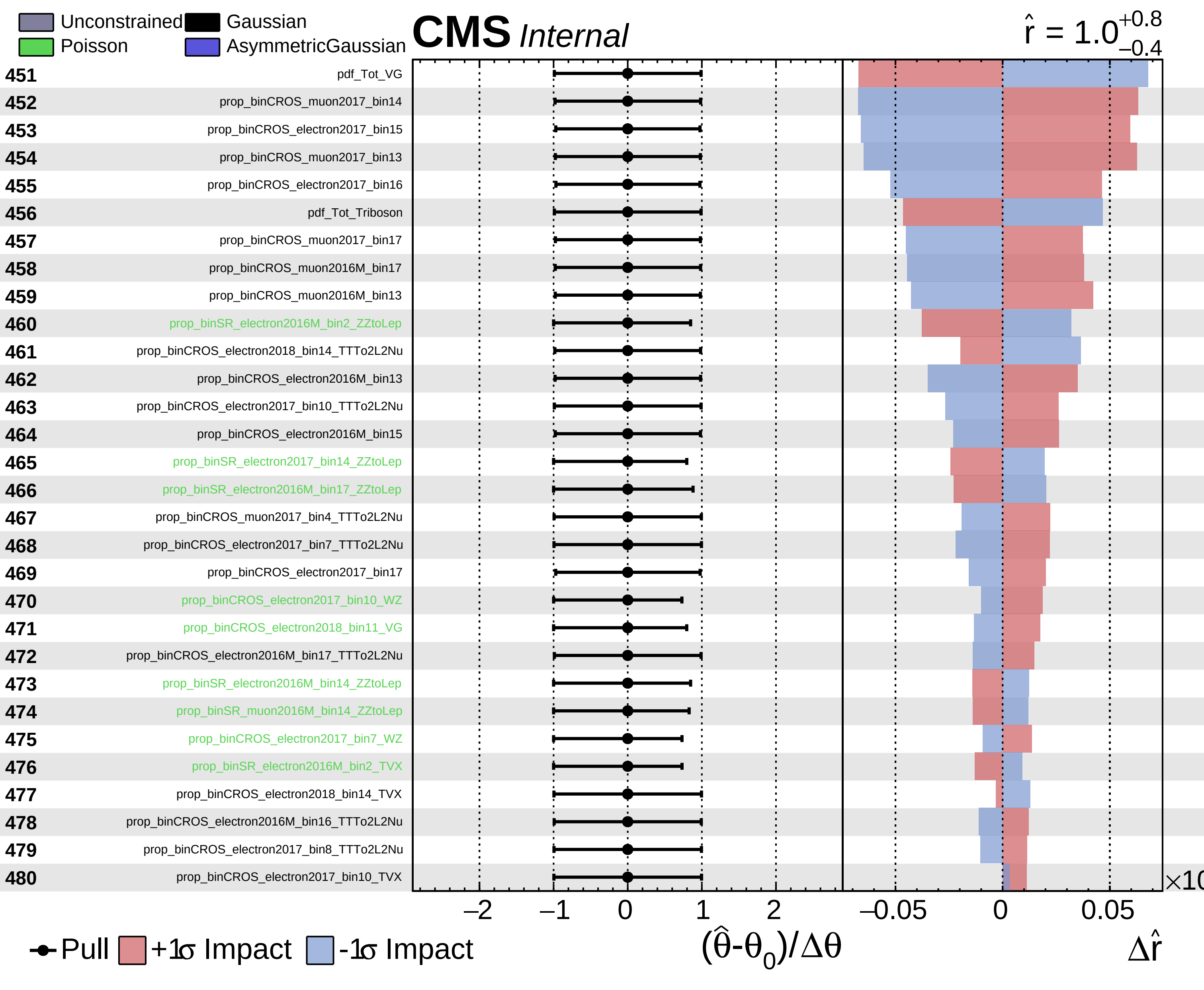


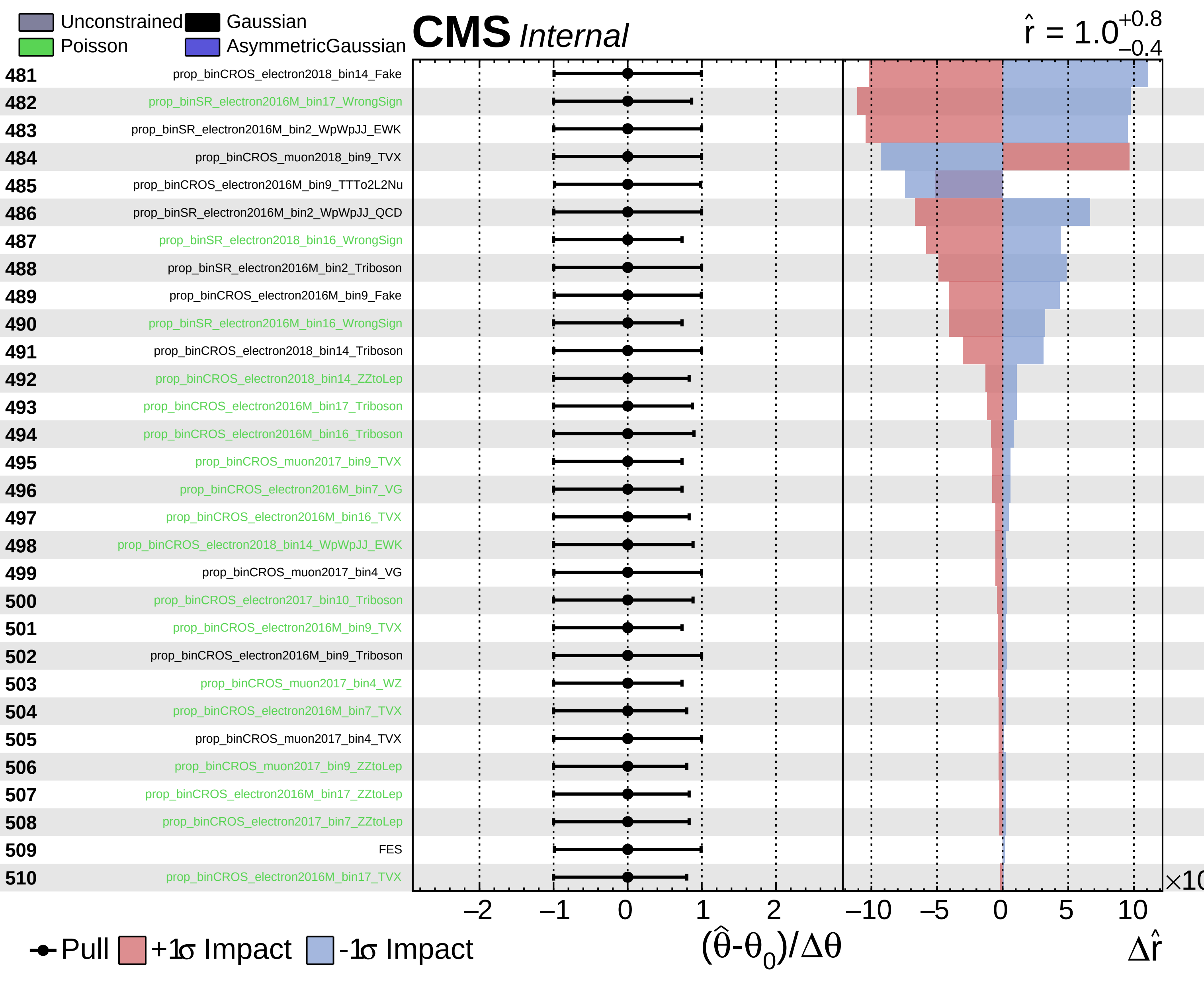
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
361 prop_binCROS_muon2017_bin6
362 prop_binCRTT_electron2016M_bin10
363 prop_binCRTT_electron2016M_bin11
364 prop_binCRTT_muon2018_bin17
365 prop_binCROS_electron2016M_bin11
366 prop_binCRTT_electron2018_bin17
367 prop_binCRTT_muon2018_bin12
368 prop_binCRTT_muon2016M_bin14
369 prop_binCRTT_muon2017_bin15
370 prop_binCROS_electron2016M_bin12
371 prop_binSR_muon2017_bin14_WpWpJJ_QCD
372 prop_binCRTT_muon2017_bin16
373 prop_binSR_electron2017_bin14_TVX
374 prop_binCRTT_electron2016M_bin8
375 prop_binCRTT_muon2016M_bin12
376 prop_binCROS_electron2018_bin15
377 prop_binCROS_muon2017_bin9_Fake
378 prop_binCRTT_electron2017_bin10
379 prop_binSR_muon2017_bin17_WpWpJJ_QCD
380 prop_binCRTT_electron2016M_bin15
381 prop_binSR_muon2017_bin17_TVX
382 prop_binCROS_muon2016M_bin11
383 prop_binCRTT_electron2016M_bin16
384 prop_binCROS_muon2017_bin15
385 prop_binSR_electron2016M_bin14_TVX
386 ISR_TTdilep
387 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
388 prop_binCRTT_electron2016M_bin14
389 prop_binCRTT_electron2017_bin15
390 prop_binCROS_electron2017_bin10_Fake
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
391 prop_binSR_electron2017_bin14_WpWpJJ_QCD
392 prop_binCRTT_electron2018_bin14
393 prop_binSR_electron2018_bin16_WpWpJJ_QCD
394 prop_binSR_electron2018_bin16_Triboson
395 prop_binCRTT_muon2016M_bin11
396 prop_binCROS_muon2017_bin16
397 prop_binSR_electron2016M_bin14_WpWpJJ_QCD
398 lep_2016M
399 prop_binCROS_electron2016M_bin17_WrongSign
400 prop_binSR_electron2017_bin8_WpWpJJ_QCD
401 prop_binCROS_muon2016M_bin16
402 prop_binSR_muon2017_bin17_Triboson
403 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
404 prop_binSR_muon2016M_bin14_Triboson
405 prop_binSR_muon2017_bin14_Triboson
406 prop_binCROS_muon2018_bin16
407 prop_binCROS_electron2017_bin11
408 prop_binCRTT_muon2017_bin9
409 prop_binSR_muon2016M_bin13_TVX
410 prop_binSR_electron2016M_bin14_WrongSign
411 prop_binCRTT_electron2017_bin12
412 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
413 prop_binCRTT_muon2016M_bin16
414 prop_binCRTT_electron2016M_bin12
415 prop_binSR_electron2016M_bin14_Triboson
416 prop_binSR_electron2017_bin8_TVX
417 prop_binSR_electron2016M_bin17_Triboson
418 prop_binSR_muon2016M_bin14_TVX
419 prop_binCROS_muon2016M_bin12
420 prop_binCRTT_electron2017_bin16
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

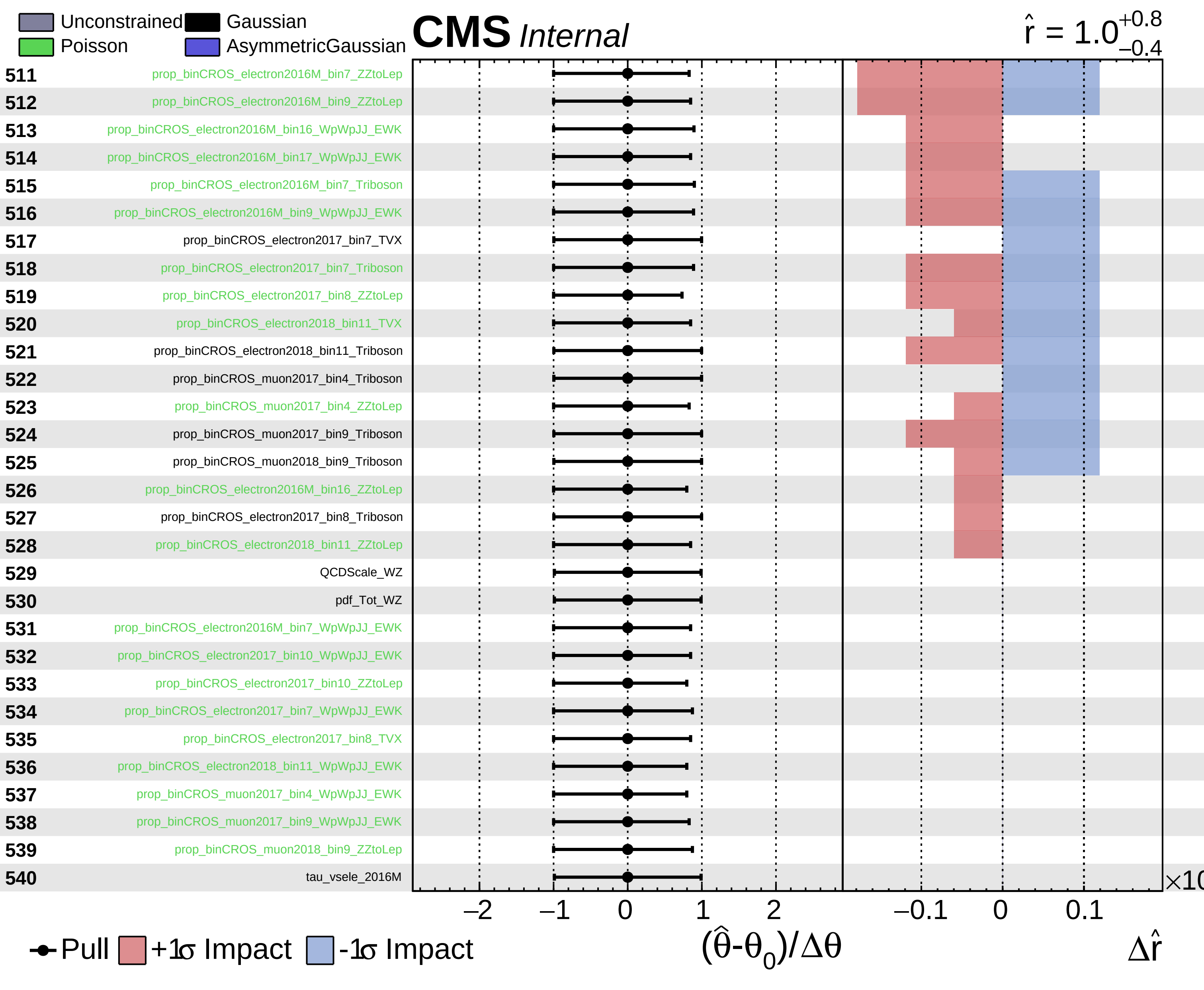


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
421 prop_binSR_electron2016M_bin16_TVX
422 prop_binCRTT_electron2016M_bin4
423 prop_binSR_electron2016M_bin16_Triboson
424 prop_binSR_muon2016M_bin13_WrongSign
425 prop_binSR_electron2017_bin14_WrongSign
426 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
427 prop_binCROS_electron2016M_bin14
428 prop_binCROS_electron2018_bin11_TTTo2L2Nu
429 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
430 prop_binSR_muon2016M_bin13_Triboson
431 prop_binCRTT_muon2016M_bin4
432 prop_binCRTT_muon2017_bin13
433 prop_binSR_electron2016M_bin2_WZ
434 prop_binSR_muon2016M_bin14_WrongSign
435 prop_binCRTT_electron2017_bin6
436 prop_binCROS_muon2017_bin9_TTTo2L2Nu
437 prop_binSR_electron2017_bin14_Triboson
438 prop_binCRTT_electron2017_bin17
439 prop_binSR_muon2017_bin17_WrongSign
440 prop_binCROS_muon2018_bin9_Fake
441 prop_binSR_electron2016M_bin2_TTTo2L2Nu
442 prop_binCRTT_muon2017_bin17
443 prop_binCRTT_muon2017_bin4
444 prop_binCROS_muon2018_bin17
445 prop_binCRTT_muon2016M_bin15
446 prop_binCRTT_muon2016M_bin17
447 prop_binSR_electron2016M_bin2_VG
448 prop_binCRTT_electron2016M_bin17
449 prop_binCROS_muon2016M_bin15
450 prop_binCROS_muon2018_bin9_TTTo2L2Nu
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

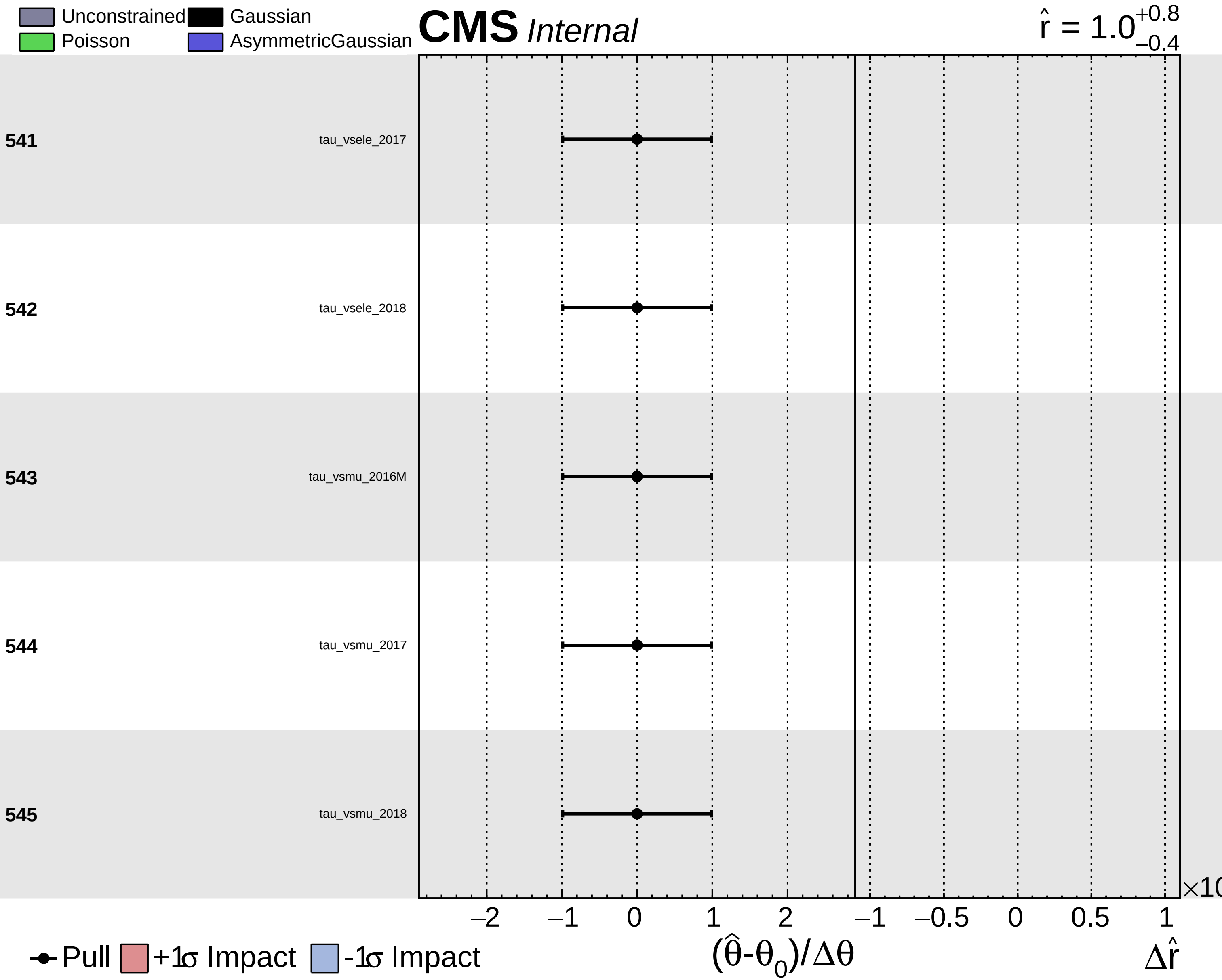

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
451 pdf_Tot_VG
452 prop_binCROS_muon2017_bin14
453 prop_binCROS_electron2017_bin15
454 prop_binCROS_muon2017_bin13
455 prop_binCROS_electron2017_bin16
456 pdf_Tot_Triboson
457 prop_binCROS_muon2017_bin17
458 prop_binCROS_muon2016M_bin17
459 prop_binCROS_muon2016M_bin13
460 prop_binSR_electron2016M_bin2_ZZtoLep
461 prop_binCROS_electron2018_bin14_TTTo2L2Nu
462 prop_binCROS_electron2016M_bin13
463 prop_binCROS_electron2017_bin10_TTTo2L2Nu
464 prop_binCROS_electron2016M_bin15
465 prop_binSR_electron2017_bin14_ZZtoLep
466 prop_binSR_electron2016M_bin17_ZZtoLep
467 prop_binCROS_muon2017_bin4_TTTo2L2Nu
468 prop_binCROS_electron2017_bin7_TTTo2L2Nu
469 prop_binCROS_electron2017_bin17
470 prop_binCROS_electron2017_bin10_WZ
471 prop_binCROS_electron2018_bin11_VG
472 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
473 prop_binSR_electron2016M_bin14_ZZtoLep
474 prop_binSR_muon2016M_bin14_ZZtoLep
475 prop_binCROS_electron2017_bin7_WZ
476 prop_binSR_electron2016M_bin2_TVX
477 prop_binCROS_electron2018_bin14_TVX
478 prop_binCROS_electron2016M_bin16_TTTo2L2Nu
479 prop_binCROS_electron2017_bin8_TTTo2L2Nu
480 prop_binCROS_electron2017_bin10_TVX
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
481 prop_binCROS_electron2018_bin14_Fake
482 prop_binSR_electron2016M_bin17_WrongSign
483 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
484 prop_binCROS_muon2018_bin9_TVX
485 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
486 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
487 prop_binSR_electron2018_bin16_WrongSign
488 prop_binSR_electron2016M_bin2_Triboson
489 prop_binCROS_electron2016M_bin9_Fake
490 prop_binSR_electron2016M_bin16_WrongSign
491 prop_binCROS_electron2018_bin14_Triboson
492 prop_binCROS_electron2018_bin14_ZZtoLep
493 prop_binCROS_electron2016M_bin17_Triboson
494 prop_binCROS_electron2016M_bin16_Triboson
495 prop_binCROS_muon2017_bin9_TVX
496 prop_binCROS_electron2016M_bin7_VG
497 prop_binCROS_electron2016M_bin16_TVX
498 prop_binCROS_electron2018_bin14_WpWpJJ_EWK
499 prop_binCROS_muon2017_bin4_VG
500 prop_binCROS_electron2017_bin10_Triboson
501 prop_binCROS_electron2016M_bin9_TVX
502 prop_binCROS_electron2016M_bin9_Triboson
503 prop_binCROS_muon2017_bin4_WZ
504 prop_binCROS_electron2016M_bin7_TVX
505 prop_binCROS_muon2017_bin4_TVX
506 prop_binCROS_muon2017_bin9_ZZtoLep
507 prop_binCROS_electron2016M_bin17_ZZtoLep
508 prop_binCROS_electron2017_bin7_ZZtoLep
509 FES
510 prop_binCROS_electron2016M_bin17_TVX
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
511 prop_binCROS_electron2016M_bin7_ZZtoLep
512 prop_binCROS_electron2016M_bin9_ZZtoLep
513 prop_binCROS_electron2016M_bin16_WpWpJJ_EWK
514 prop_binCROS_electron2016M_bin17_WpWpJJ_EWK
515 prop_binCROS_electron2016M_bin7_Triboson
516 prop_binCROS_electron2016M_bin9_WpWpJJ_EWK
517 prop_binCROS_electron2017_bin7_TVX
518 prop_binCROS_electron2017_bin7_Triboson
519 prop_binCROS_electron2017_bin8_ZZtoLep
520 prop_binCROS_electron2018_bin11_TVX
521 prop_binCROS_electron2018_bin11_Triboson
522 prop_binCROS_muon2017_bin4_Triboson
523 prop_binCROS_muon2017_bin4_ZZtoLep
524 prop_binCROS_muon2017_bin9_Triboson
525 prop_binCROS_muon2018_bin9_Triboson
526 prop_binCROS_electron2016M_bin16_ZZtoLep
527 prop_binCROS_electron2017_bin8_Triboson
528 prop_binCROS_electron2018_bin11_ZZtoLep
529 QCDScale_WZ
530 pdf_Tot_WZ
531 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
532 prop_binCROS_electron2017_bin10_WpWpJJ_EWK
533 prop_binCROS_electron2017_bin10_ZZtoLep
534 prop_binCROS_electron2017_bin7_WpWpJJ_EWK
535 prop_binCROS_electron2017_bin8_TVX
536 prop_binCROS_electron2018_bin11_WpWpJJ_EWK
537 prop_binCROS_muon2017_bin4_WpWpJJ_EWK
538 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
539 prop_binCROS_muon2018_bin9_ZZtoLep
540 tau_vsele_2016M
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
541
tau_vsele_2017
542
tau_vsele_2018
543
tau_vsmu_2016M
544
tau_vsmu_2017
545
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$